

IAN NASTAJUS

3D SCIENTIFIC VISUALIZATION | TECHNICAL
ILLUSTRATION | BLENDER

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PROFILE

Technical 3D visualizer with a computer-science foundation and Blender experience translating complex scientific and engineering concepts into clear, publication-oriented imagery. Contributed 3D scientific visualizations to two peer-reviewed 2026 journal publications. Strongest in structured hard-surface modeling, repeated/layered geometry, materials and lighting, camera composition, and iterative visual problem-solving.

SELECTED 3D / SCIENTIFIC VISUALIZATION ACCOMPLISHMENTS

- Created the central Blender-based plasmonic metamaterial thermal-emitter model and integrated it into a completed scientific graphic published in Frontiers of Optoelectronics (2026), combining 3D structure, spectral plots, measurement-plane annotations and visual effects.
- Created 3D scientific renderings for a «Cu/TiO2 metasurface» absorber and fabrication-process visualization published in Advanced Optical Materials (2026), depicting layered materials, periodic meta-atoms and sequential fabrication stages.
- Built repeated nanoscale-style geometry using arrays, bevels, linked materials/meshes, precise placement, scene scaling, camera framing and HD render setup.
- Created optical-wave / beam visualization studies using curves, emissive materials, transparency, lighting, camera controls and spatial composition.
- Produced material and colour studies for metasurface stacks and nanopillar/nanopost variants using Principled BSDF, metallic/roughness contrast and controlled multi-light setups.
- Worked iteratively from incomplete scientific references and researcher feedback, translating unfamiliar concepts into readable visual components while preserving a clear boundary between visualization work and scientific authorship.

CORE SKILLS

BLENDER / 3D	VISUAL / TECHNICAL COMMUNICATION
• Hard-surface & schematic modeling • Arrays, booleans, bevels, extrusion, curves • Modifiers, snapping, hierarchy & linked data • Principled materials; metallic / roughness / emission • Lighting, shadows, camera framing & render setup	• Scientific & technical schematics • Metasurface / nanoscale conceptual structures • Process and fabrication visualization • Colour, material & visual-hierarchy iteration • Figure composition & stakeholder iteration

TECHNICAL FOUNDATION

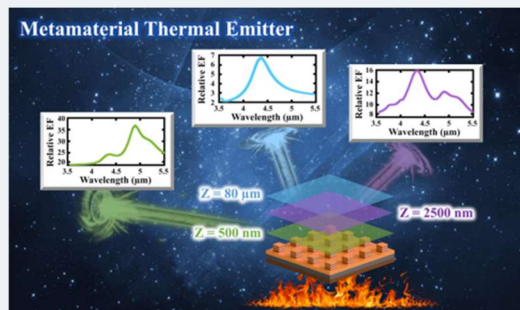
B.Sc. Computer Science, Toronto Metropolitan University (Ryerson), 2014

Earlier analyst-programming, software-development and game-development experience; familiarity with Unity and technical production workflows.

SELECTED 3D PORTFOLIO

Two published scientific-visualization contributions followed by two supporting Blender studies.

01 › PUBLISHED: METAMATERIAL THERMAL EMITTER



Created the central Blender thermal-emitter model and integrated it with provided MATLAB plots, observation planes, labels and visual effects in PowerPoint for the final published figure.

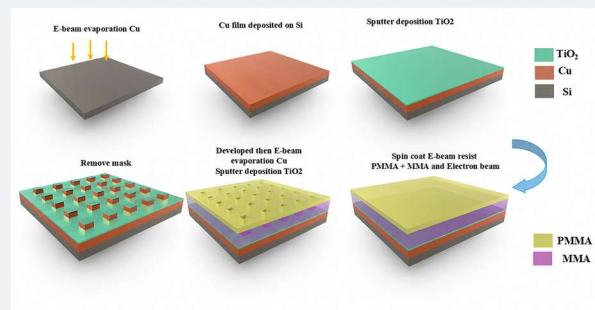
PEER-REVIEWED JOURNAL ARTICLE

Frontiers of Optoelectronics (2026)

<https://journal.hep.com.cn/foe/EN/10.2738/foe.2026.0032>

Demonstrates: scientific 3D illustration · metasurface modeling · figure composition · materials & lighting · 2D/3D integration

02 › PUBLISHED: Cu/TiO₂ METASURFACE ABSORBER



Created the individual Blender renders depicting stages of Cu/TiO₂ metasurface fabrication, including substrate layers, thin-film deposition, resist layers and patterned structures.

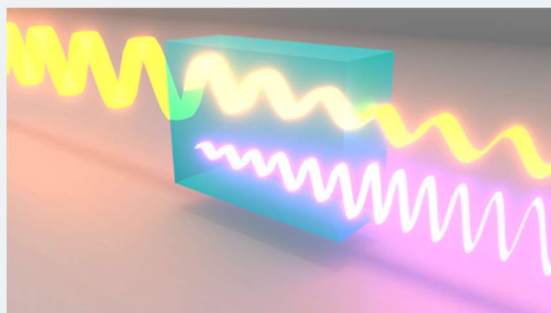
PEER-REVIEWED JOURNAL ARTICLE

Advanced Optical Materials (2026)

<https://advanced.onlinelibrary.wiley.com/doi/full/10.1002/adom.71514>

Demonstrates: arrays · hard-surface modeling · layered geometry · material assignment · transparency

03 › OPTICAL WAVE / BEAM STUDY



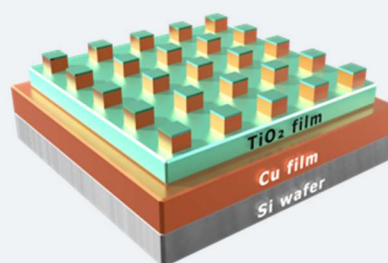
Blender study of emissive optical-wave paths interacting with a transparent volume, combining glow, transmission, lighting and compositing for spatial readability.

SUPPORTING BLENDER PORTFOLIO STUDY

Optical / emissive-material visualization

Demonstrates: curves · emission · transparency · multi-light balancing · render compositing · camera composition

04 › LABELED METASURFACE STACK



Clean technical render of a Cu/TiO₂ metasurface stack, emphasizing periodic geometry, reflective material response, controlled illumination and isolated light influence.

SUPPORTING BLENDER PORTFOLIO STUDY

Technical metasurface rendering study

Demonstrates: arrays · layered geometry · reflectivity · material tuning · light linking / isolation